

Practical -10

1)

```
bharath_sai_reddy@LAPTOP-1 × + ▾
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical$ cd Practical-10
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/Practical-10$ echo "Hello OSSP" > original.txt
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/Practical-10$ ls -li original.txt
1472 original.txt
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/Practical-10$ stat original.txt
File: original.txt
  Size: 11          Blocks: 8          IO Block: 4096   regular file
Device: 8,48      Inode: 1472           Links: 1
Access: (0644/-rw-r--r--)  Uid: ( 1000/bharath_sai_reddy)   Gid: ( 1000/bharath_sai_reddy)
Access: 2026-10-01 09:45:34.210274079 +0000
Modify: 2026-10-01 09:45:34.210274079 +0000
Change: 2026-10-01 09:45:34.210274079 +0000
Birth: 2026-10-01 09:45:34.210274079 +0000
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/Practical-10$ ln original.txt hardlink.txt
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/Practical-10$ ls -li
total 8
1472 -rw-r--r-- 2 bharath_sai_reddy bharath_sai_reddy 11 Oct  1 09:45 hardlink.txt
1472 -rw-r--r-- 2 bharath_sai_reddy bharath_sai_reddy 11 Oct  1 09:45 original.txt
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/Practical-10$ ln -s original.txt symlink.txt
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/Practical-10$ ls -li
total 8
1472 -rw-r--r-- 2 bharath_sai_reddy bharath_sai_reddy 11 Oct  1 09:45 hardlink.txt
1472 -rw-r--r-- 2 bharath_sai_reddy bharath_sai_reddy 11 Oct  1 09:45 original.txt
12270 lrwxrwxrwx 1 bharath_sai_reddy bharath_sai_reddy 12 Oct  1 09:50 symlink.txt -> original.txt
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/Practical-10$
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2) Mmap_file

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bharath_sai_reddy@LAPTOP-1 × + ▾
GNU nano 7.2                                     mmap_file.c *
#include <stdio.h>
#include <fcntl.h>
#include <sys/mman.h>
#include <sys/stat.h>
#include <unistd.h>
#include <string.h>

int main() {
    int fd = open("data.txt", O_RDWR | O_CREAT, 0666);

    const char *msg = "Hello using mmap!";
    write(fd, msg, strlen(msg));

    struct stat st;
    fstat(fd, &st);

    char *data = mmap(NULL, st.st_size,
                      PROT_READ | PROT_WRITE,
                      MAP_SHARED, fd, 0);

    if (data == MAP_FAILED) {
        perror("mmap");
        return 1;
    }

    printf("File content: %s\n", data);

    munmap(data, st.st_size);
    close(fd);

    return 0;
}
```

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bharath_sai_reddy@LAPTOP-1  ×  +  ▾  
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/Practical-10$ ./mmap_file  
File content: Hello using mmap!  
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/Practical-10$ cat data.txt  
Hello using mmap!bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Practical/Practical-10$
```